Skills & Orchestration Layer - Training Slides

Slide 1: The Problem — Why Chat-Style AI Coding Falls Short

Level 0: Raw Prompting (What most devs do today)

```
You:   "Add user authentication"
AI:    Here's a basic auth implementation... [shallow code snippet]
You:   "No, I need JWT tokens"
AI:    Here's JWT... [another snippet, no context from before]
You:   "It doesn't work with my existing User model"
AI:    Can you paste your User model? [lost all prior context]
You:   [pastes 200 lines] "Now fix the middleware too"
AI:    [hallucinated middleware that doesn't match your codebase]
...
45 minutes later: scattered code, no tests, no docs, manual copy-paste
```

Pain points:

- **Shallow work** — AI gives generic snippets, not production code
- **No codebase awareness** — doesn't know your files, patterns, conventions
- **Context evaporates** — each message starts from scratch
- **Manual everything** — you copy-paste, test, review, commit yourself
- **No quality gates** — no one checks the AI's output before you ship it

Level 1: Skills (Better, but still fragmented)

Skills give AI **domain expertise** — how to write backends, style UIs, test properly. But standalone skills are isolated. Each does one thing well in a vacuum:

```
/skill-backend    → generates API code      ...but who plans it?
/skill-test       → writes test cases       ...but who runs them?
/skill-review     → reviews code quality    ...but who fixes issues?
/skill-git        → commits changes         ...but who coordinates?
```

Skills without orchestration = **a toolbox with no builder**. You still manually chain them.

Level 2: Orchestration (The missing layer)

What if one command could **plan, code, test, review, and ship** — automatically?

```
/ck-cook "Add user auth"
  │
  ├─ 1. Scout      → scout agent scans codebase, finds relevant files
  ├─ 2. Research    → researcher agents gather information (parallel)
  ├─ 3. Plan        → planner agent designs architecture & analysis
  ├─ 4. Implement   → fullstack-developer agent writes code (parallel moc
  ├─ 5. Simplify    → code-simplifier agent refines code
  ├─ 6. Test        → tester agent runs tests, debugger agent if failures
  ├─ 7. Review      → code-reviewer agent evaluates quality
  └─ 8. Finalize    → docs-manager updates docs, git-manager commits
```

10 agents orchestrated by **1 command**. The rest is handled automatically.

The Progression

```
┌──────────────┐     ┌──────────────┐     ┌──────────────────────┐
│  Level 0     │     │  Level 1     │     │  Level 2             │
│  RAW CHAT    │ ──> │  SKILLS      │ ──> │  ORCHESTRATION       │
│              │     │              │     │                      │
│ • Shallow    │     │ • Deep but   │     │ • Deep AND           │
│ • No context│     │   isolated   │     │   coordinated        │
│ • Manual     │     │ • Manual     │     │ • Automatic           │
│ • No gates   │     │   chaining   │     │ • Quality gates      │
│              │     │              │     │ • 10 agents, 1 cmd   │
│ 45 min/feat │     │ 20 min/feat │     │ 5 min/feat           │
└──────────────┘     └──────────────┘     └──────────────────────┘
```

CoKit operates at Level 2. That's the difference.

Slide 2: What Is the Orchestration Layer?

The system that **coordinates AI agents, prompts, and skills** to deliver end-to-end development workflows.

```
┌──────────────────────────────────────────────────────┐
│            USER: /ck-cook "Add user auth"            │
└──────────────────────────────────────────────────────┘
                  │
                  ▼
┌──────────────────────────────────────────────────────┐
│        ORCHESTRATION LAYER (4 Tiers)                 │
│                                                      │
│  Tier 1 → Tier 2 → Tier 3 → Tier 4                  │
│  (command) (execute) (tools) (domain knowledge)      │
│                                                      │
```

```
                ▼
┌──────────────────────────────────────────────────────┐
│              DELIVERED FEATURE                       │
│     Code + Tests + Docs + Commit                     │
└──────────────────────────────────────────────────────┘
```

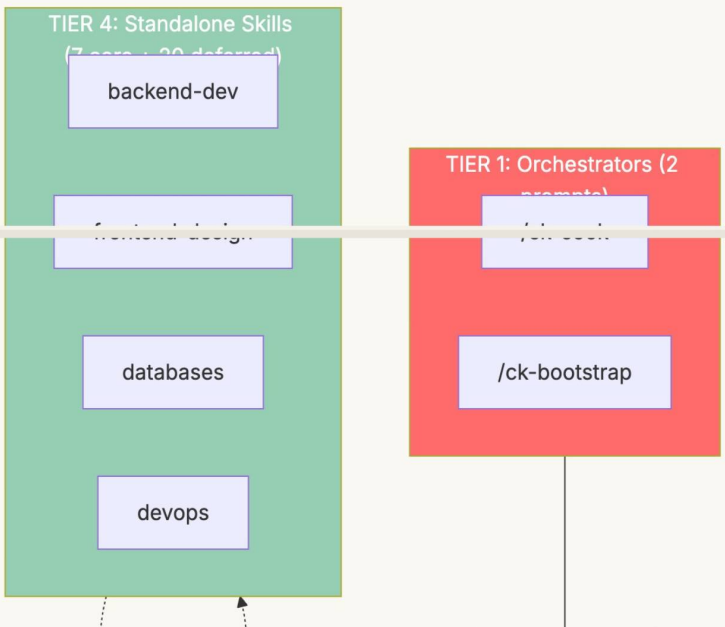
Without orchestration → you prompt one step at a time.
With orchestration → 1 command triggers the full pipeline.

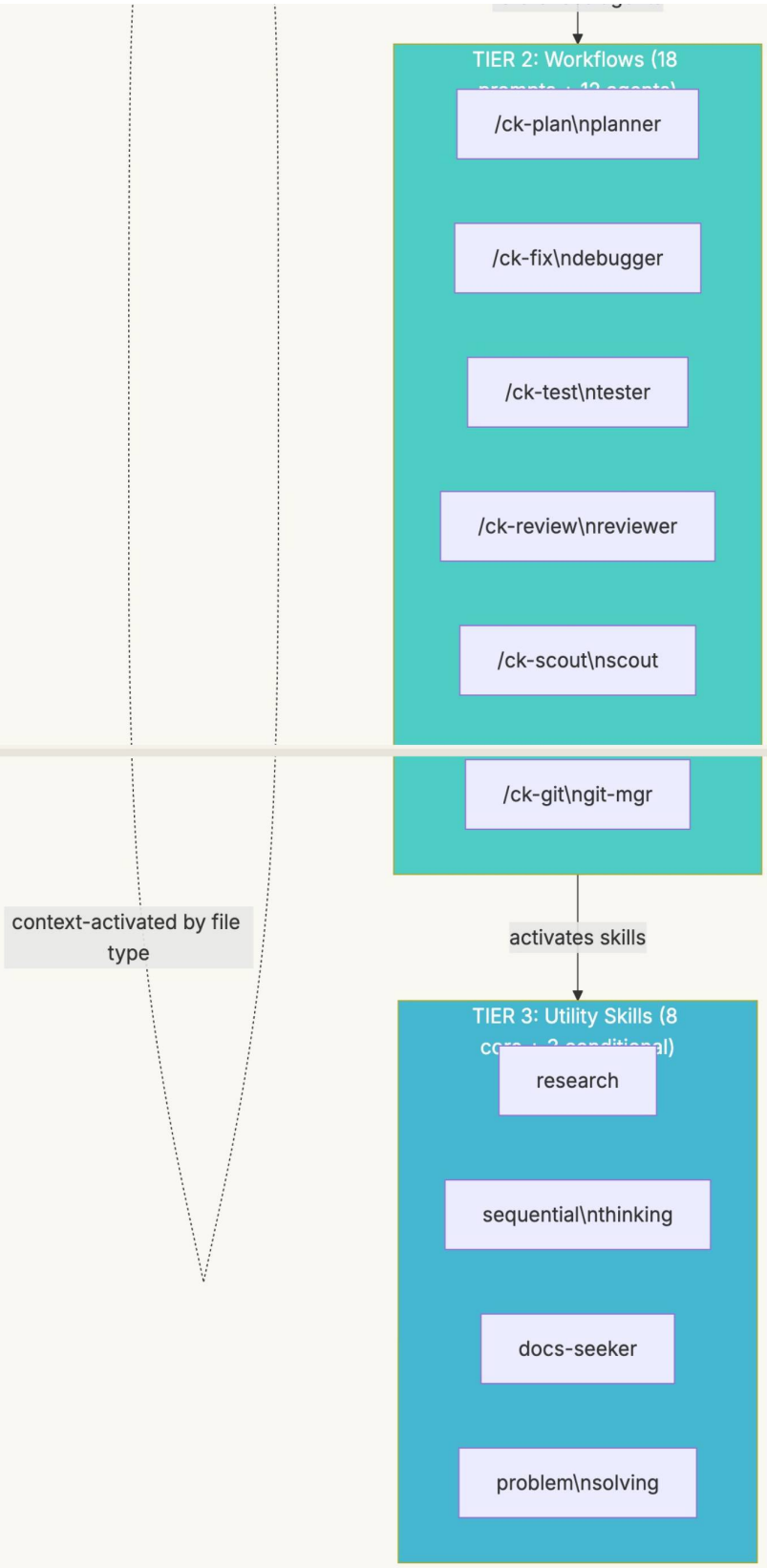
Slide 3: The Numbers at a Glance

Resource	Count	Purpose
Tiers	**4**	Layered hub-and-spoke architecture
Prompts	**34**	Command templates (18 core + 10 variants + 6 spec)
Agents	**12**	Specialized AI personas
Skills	**27**	Domain expertise packages
Instructions	**5**	Context-aware rules auto-loaded by file type
Collections	**5**	Bundled resource groups
Rules	**4**	Orchestration protocol files

Total: 87 resources across 4 tiers.

Slide 4: 4-Tier Architecture Overview





Rule: Each tier only calls DOWN. No circular dependencies. No upward calls.

Tier	Role	Answers
1	End-to-end workflow controllers	**WHAT** to do and in what order
2	Domain-specific executors	**HOW** to do each step
3	Reusable capability providers	**WITH WHAT** methodology
4	Self-contained domain experts	**ABOUT WHAT** domain

Slide 5: Tier 1 — Orchestrators

2 prompts | End-to-end workflow controllers

Prompt	What it does	Agents it references
`/ck-cook`	Full pipeline: Scout → Research → Plan → Implement → Simplify → Test → Review → Finalize	researcher, scout, planner, ui-ux-designer, fullstack-developer, code-simplifier, tester, debugger, code-reviewer, docs-manager, git-manager
`/ck-bootstrap`	New project scaffolding	researcher, planner

Think of it as: The project manager that knows all steps and delegates to specialists.

Cook has **6 modes** based on input detection:

Mode	Trigger	Behavior
interactive	default	User approval at each gate
auto	"auto", "trust me"	Auto-approve if score ≥ 9.5
fast	"fast", "quick"	Skip research phase
parallel	"parallel", 3+ features	Multi-agent execution
no-test	"no test", "skip test"	Skip testing step
code	path to plan.md	Execute existing plan

Slide 6: Tier 2 — Workflow Prompts + Agents

18 prompts + 12 agents | Each command = **Prompt** (workflow) + **Agent** (persona)

Category	Commands	Agent	Skills activated
Planning	`/ck-plan` , `-fast` , `-hard` , `-validate` , `-red-team`	`planner`	planning, research, docs-seeker
Scouting	`/ck-scout`	`scout`	repomix (if available)
Debugging	`/ck-debug`	`debugger`	debug methodology (4 phases)

Fixing	`/ck-fix` , `-types` , `-test` , `-ci` , `-ui` , `-fast` , `-hard` , `-logs`	`debugger`	debug, problem-solving, sequential-thinking
Testing	`/ck-test`	`tester`	web-testing
Review	`/ck-review`	`code-reviewer`	sequential-thinking
Brainstorm	`/ck-brainstorm`	`brainstormer`	research, docs-seeker
Simplify	`/ck-simplify`	`code-simplifier`	—
Git	`/ck-git`	`git-manager`	git
Docs	`/ck-docs`	`docs-manager`	docs-seeker
Q&A	`/ck-ask` , `/ck-watzup`	(none)	context-dependent

12 agents serve **18 prompts** — `debugger` alone powers 9 prompt variants.

Prompt Navigation (Not Cross-Hub Orchestration)

Tier 2 prompts link to each other via **"Suggested Next Steps"** footers:

```
/ck-fix → suggests → /ck-test, /ck-git
/ck-test → suggests → /ck-fix, /ck-review, /ck-git
/ck-brainstorm → suggests → /ck-plan (step 8: "Create implementation plar
/ck-review → suggests → /ck-fix, /ck-git
```

This is **sequential prompt chaining** (user chooses next step), NOT automatic cross-hub orchestration. The user decides whether to follow suggested navigation.

Slide 7: Tier 3 — Utility Skills

8 core + 3 conditional | Reusable, stateless capability providers

```
                  ┌──────────────────┐
     ┌──────────>│ sequential-      │<──────────┐
     │            │ thinking         │           │
     │            └──────────────────┘           │
     │                   ▲                       │
┌────┴─────┐      ┌──────┴──────────┐      ┌────┴─────┐
│ debugger │      │ code-reviewer   │      │ planner  │
└──────────┘      └─────────────────┘      └──────────┘
```

```
→ Leaf nodes: never call other skills
```

Skill	Capability	Referenced by
research	Web search, docs synthesis	planner, brainstormer
docs-seeker	Library docs via context7	planner, brainstormer, docs-manager
sequential-thinking	Step-by-step reasoning	debugger, planner, code-reviewer, brainstormer
problem-solving	Systematic stuck-ness techniques	debugger (fix skill)
agent-browser	Browser automation CLI	debugger (fix-ui), tester
repomix *(if available)*	Pack repo for AI	scout
context-engineering	Token optimization	any agent when constrained
mermaidjs-v11	Diagram syntax v11	planner, visual generation
ui-ux-pro-max *(if available)*	Design intelligence	ui-ux-designer
ai-multimodal *(if available)*	Image/video analysis (Gemini)	any agent needing vision
media-processing *(if available)*	FFmpeg/ImageMagick	any agent needing media ops

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Slide 8: Tier 4 — Standalone Skills

7 core + 20 deferred | Self-contained domain experts, no CoKit core dependency

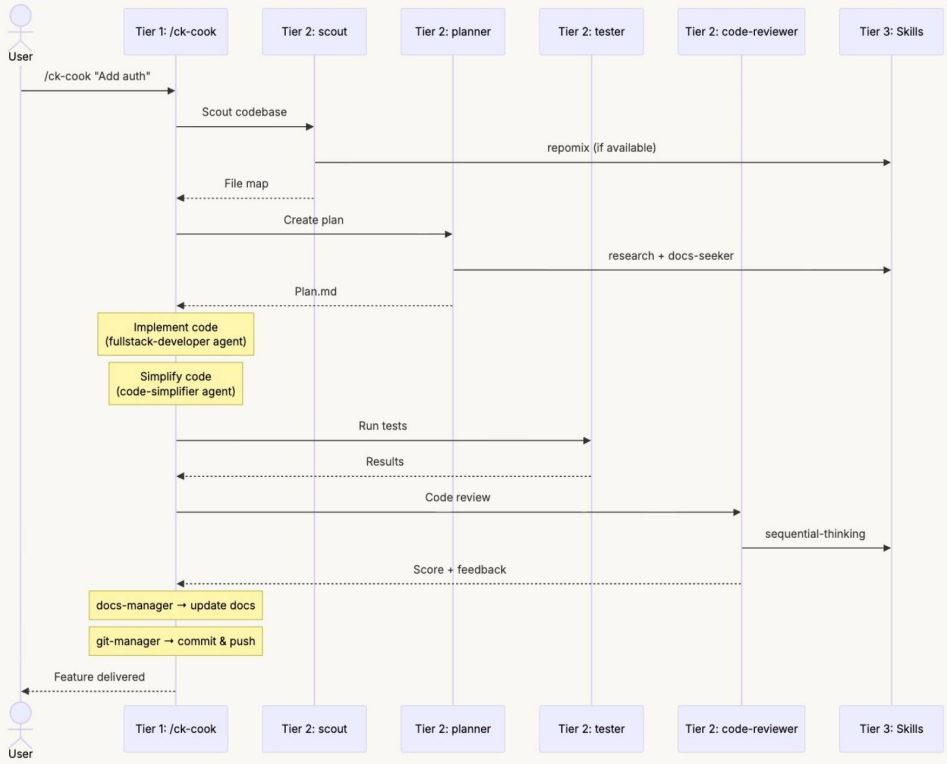
Category	Skills
Backend	`backend-development`, `databases`, `devops`
Frontend	`frontend-design`, `ui-styling`
Testing	`web-testing`
Git	`git`
MCP	`mcp-management`

How they differ from Tier 3:

- Tier 3: **Methodology** — how to think, research, debug
- Tier 4: **Domain knowledge** — how to build backends, style UIs, deploy

Pluggable: Add `skills/ck-new-skill/SKILL.md` → auto-discovered by context. No changes elsewhere.

Slide 9: The Full Communication Flow



Slide 10: Design Principles

1. Hub-and-Spoke (Not Mesh)

Orchestrators (hub) delegate to specialists (spokes). **No peer-to-peer** agent communication. Tier 2 prompts suggest next steps to each other, but the **user decides** — it's navigation, not automatic orchestration.

2. Separation of Concerns

Each resource does ONE thing well:

- **Prompt** = defines a workflow
- **Agent** = defines expertise/persona
- **Skill** = provides knowledge/methodology

3. DRY via Shared Skills

`sequential-thinking` written once → used by debugger, planner, code-reviewer, brainstormer.

4. Graceful Degradation

External tools use `(if available)`. Missing repomix? Scout falls back to built-in search.

5. Pluggable Architecture

New skill = new SKILL.md file. No registration. No changes to prompts. Context-activated.

Slide 11: Cheat Sheet

```
┌──────────────────────────────────────────────────────────────┐
│                 COKIT ORCHESTRATION CHEAT SHEET               │
├──────────────────────────────────────────────────────────────┤
│                                                               │
│  TIER 1 (2)    /ck-cook, /ck-bootstrap                       │
│  ─────────     Full workflow controllers                      │
│                                                               │
│  TIER 2 (30)   18 prompts + 12 agents                        │
│  ─────────     /ck-plan, /ck-fix, /ck-test, /ck-review...    │
│                Domain-specific executors                      │
│                                                               │
│  TIER 3 (11)   8 core + 3 conditional skills                 │
│  ─────────     research, sequential-thinking, docs-seeker    │
│                Reusable capability providers                  │
│                                                               │
│  TIER 4 (27)   7 core + 20 deferred skills                   │
│  ─────────     backend-dev, databases, devops, ui-styling    │
│                Self-contained domain experts                  │
```

```
│                                                          │
│  TOTAL: 4 tiers │ 34 prompts │ 12 agents │ 27 skills      │
│          5 instructions │ 5 collections │ 4 rules         │
│                                                          │
│  ONE COMMAND TO RULE THEM ALL: /ck-cook                  │
│                                                          │
└──────────────────────────────────────────────────────────┘
```

Slide 12: Try It Yourself

Want to...	Run
Build a feature end-to-end	`/ck-cook "description"`
Plan before building	`/ck-plan "description"`
Fix a bug	`/ck-fix "error message"`
Fix TypeScript errors	`/ck-fix-types`
Run & analyze tests	`/ck-test`
Review code quality	`/ck-review`
Search the codebase	`/ck-scout "what to find"`
Brainstorm solutions	`/ck-brainstorm "problem"`
Quick answer	`/ck-ask "question"`
Commit changes	`/ck-git`

Pro tip: Start with `/ck-cook` — it orchestrates everything else automatically.